

Lab 4: Network Monitoring & Security Analysis

Team Members

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Lab Overview

01

Packet Capture

Capture network traffic and analyze IP addresses, MAC addresses, and protocols.

02

Ping Flood Attack

Simulate a ping flood attack and observe network behavior.

03

Port Scanning

Detect port scanning attempts using Telnet to probe open ports on a server.

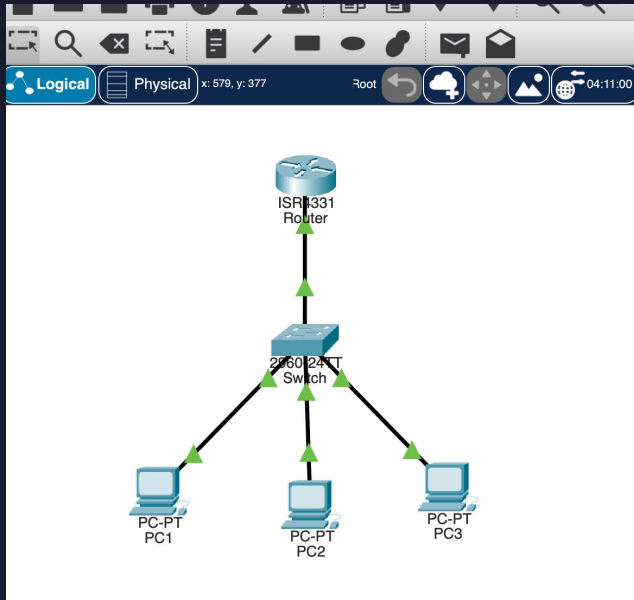
04

Traffic Anomalies

Identify traffic anomalies such as high volume and unusual packet behavior.

Network Topology

Network Topology



IP Configuration Table

PC1	Monitoring PC	192.168.1.10
PC2	Victim PC	192.168.1.20
PC3	Attacker PC	192.168.1.30
Server	Port Scan Target	192.168.1.50
Router GEO/0/0	Default Gateway	192.168.1.1

All devices were placed on the same 192.168.1.0/24 subnet with the router acting as the default gateway at 192.168.1.1. Each PC was assigned a unique static IP address. PC1 acts as a monitoring device, PC2 is the victim target, and PC3 is the attacker. The Server-PT was added later for the port scanning task.

Router Configuration

Router CLI

```

Router0
  Physical  Config  CLI  Attributes
  IOS Command Line Interface

use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wai/export/crypto/tool/stgry.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco 128433.169 (18K) processor with 17959998/4147K bytes of memory.
Processor board ID F4C3201000
1 Gigabit Ethernet interface
4749K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
327147K bytes of flash memory at bootflash:.
2K bytes of M801 ROM files at bootflash.

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
%Invalid interface type and number
Router(config)#interface GigabitEthernet0/0
%Invalid interface type and number
Router(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 unassigned YES unset administratively down down
GigabitEthernet0/0/1 unassigned YES unset administratively down down
GigabitEthernet0/0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
Router(config-if)#exit
Router(config)#exit
Router#
sys>>>CONFIG_1: Configured from console by console
Router#
  
```

Key CLI Commands

show ip interface brief

Revealed the correct interface name was GigabitEthernet0/0/0. The command GigabitEthernet0/0 returned an error

interface GigabitEthernet0/0/0

Entered interface configuration mode for the correct interface

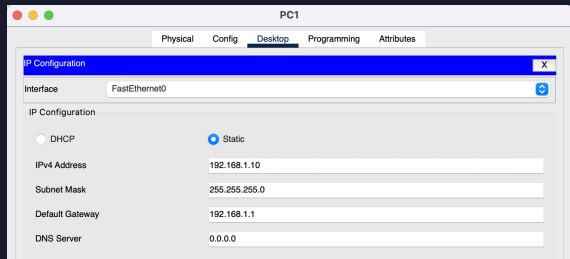
ip address 192.168.1.1 255.255.255.0

Assigned the gateway IP address to the router interface

no shutdown

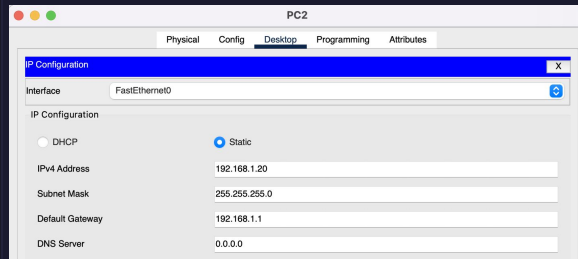
Brought the interface up. The cable between Switch and Router turned green, confirming the link was active

PC IP Configuration & Connectivity Test



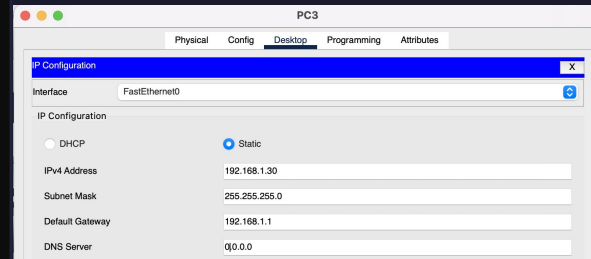
PC1 — Monitoring

192.168.1.10



PC2 — Victim

192.168.1.20



PC3 — Attacker

192.168.1.30

Each PC was manually configured with a static IP address, subnet mask of 255.255.255.0, and default gateway of 192.168.1.1. PC1 pinged PC2 at 192.168.1.20 to verify the network was fully operational before beginning any simulation tasks. All 4 packets were received with 0% packet loss, confirming every device was correctly connected through the switch and router.

4/4

0% loss



Network OK

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

PDU Packet Details

PDU Information: Inbound PDU Details

PDU Information at Device: Switch

OST Model Inbound PDU Details Outbound PDU Details

PDU Formats

Ethernet II

0	4	8	Bytes
PREAMBLE: 101010..1		DEST ADDR: FFFF.FFFF	
SRC ADDR :000A.41A0		DATA (VAR IABLE LEN	FCS: 0x0000

Arp

0	8	16	Bits
HARDWARE TYPE: 0x0001		PROTOCOL TYPE: 0x0800	
HLEN: 0x06	PLEN: 0x04	OPCODE: 0x0001	
SOURCE MAC :000A.41A0.DA41			
SOURCE IP :192.168.1.10			
TARGET MAC: 0000.0000.0000			
TARGET IP: 192.168.1.20			

Simulation Panel

Event List

Last Device

PC1

Switch

Switch

PC2

Switch

Reset Simulation ☒ Constant Delay Captured to: 1.012 s

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT, TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

Last Status Source Destination Type Color

SOURCE MAC

000A.41A0.DA41

The physical hardware address of PC1's network interface card.

SOURCE IP

192.168.1.10

PC1's IP address: the device that initiated the ping.

TARGET IP

192.168.1.20

PC2's IP address: the intended destination of the ping.

PROTOCOL

ARP (0x0806)

Before ICMP ping can be sent, ARP resolves the MAC address of the target.

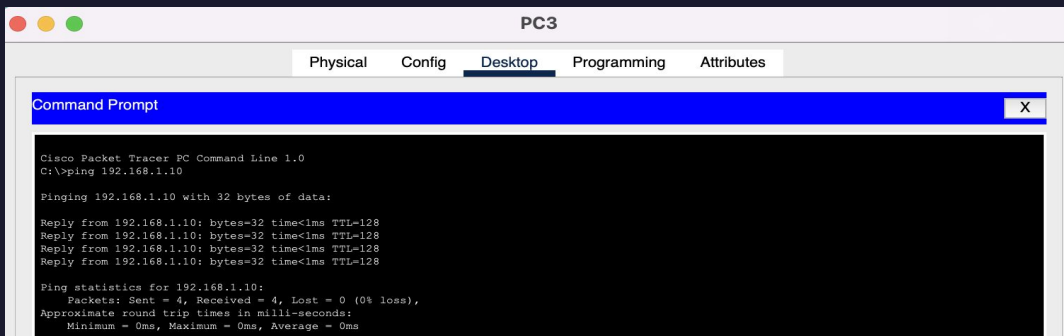
DEST MAC

FFFF.FFFF.FFFF

Broadcast MAC: ARP asks all devices on the network who has 192.168.1.20.

Simulating a DoS Attack

PC3 Command Prompt: ping 192.168.1.10



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Attack Details

Attacker: PC3 — 192.168.1.30

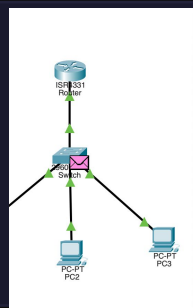
Target: PC1 — 192.168.1.10

Protocol: ICMP Echo Request

Attack Type: Denial of Service (DoS)

Detection: High-volume ICMP in Event List

From PC3's Command Prompt, we ran ping 192.168.1.10 to flood PC1 with ICMP packets. The Event List on the right shows the packets rapidly bouncing between PC3, the Switch, and PC1 in a continuous loop — this high-volume repetition is what makes it a DoS attack.

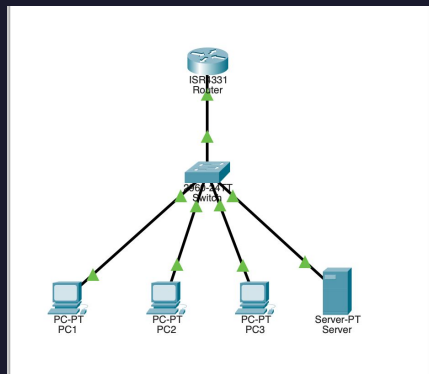


Event List	Time(sec)	Last Device	All Device	Type
Vis.	0.000	--	PC3	ICMP
	0.001	PC3	Switch	ICMP
	0.002	Switch	PC1	ICMP
	0.003	PC1	Switch	ICMP
	0.004	Switch	PC3	ICMP
	0.870	--	Switch	STP
	0.871	Switch	PC3	STP
	0.871	Switch	Router	STP
	0.871	Switch	PC1	STP
	0.871	Switch	PC2	STP
	1.008	--	PC3	ICMP
	1.009	PC3	Switch	ICMP
	1.010	Switch	PC1	ICMP
	1.011	PC1	Switch	ICMP
	1.012	Switch	PC3	ICMP
	2.016	--	PC3	ICMP
	2.016	PC3	Switch	ICMP
	2.017	Switch	PC1	ICMP
	2.018	PC1	Switch	ICMP

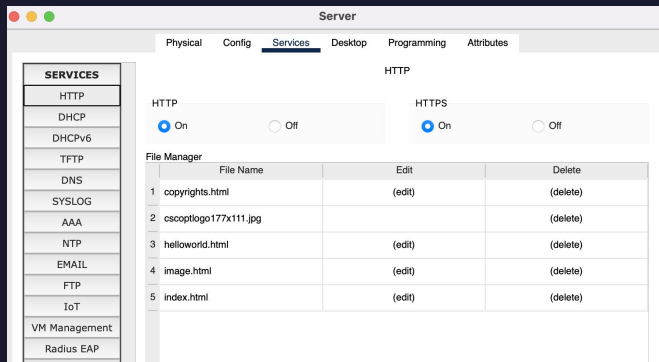
Server Setup & Open Ports

We simulated port scanning by adding a Server and using Telnet to probe individual ports, generating the same TCP traffic nmap would produce

Server added to topology at 192.168.1.50, connected to the Switch on FastEthernet0/4



Server Services tab shows HTTP (port 80) and FTP (port 21) are ON, these are the open ports a port scanner would detect



Port 80

HTTP — OPEN

Web server traffic

Port 443

HTTPS — OPEN

Secure web traffic

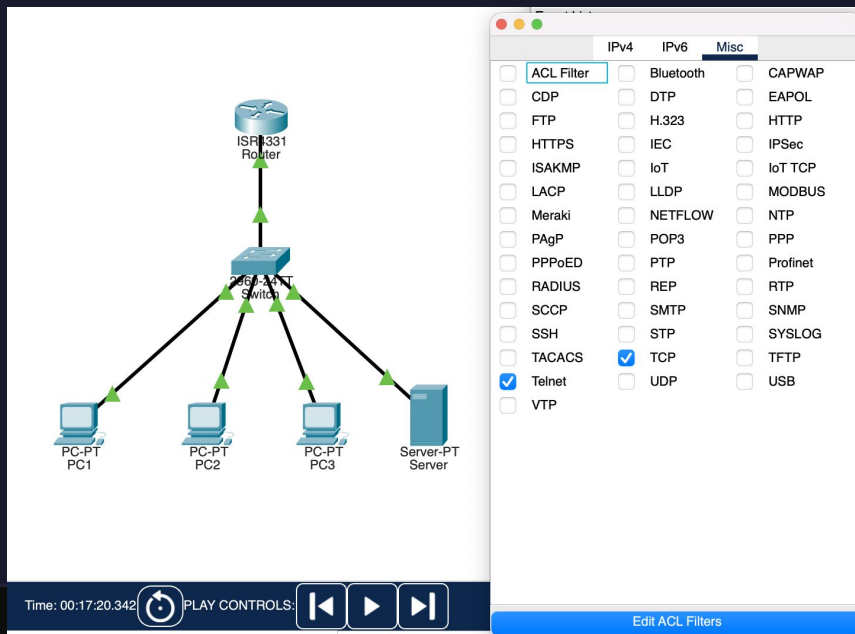
Port 21

FTP — OPEN

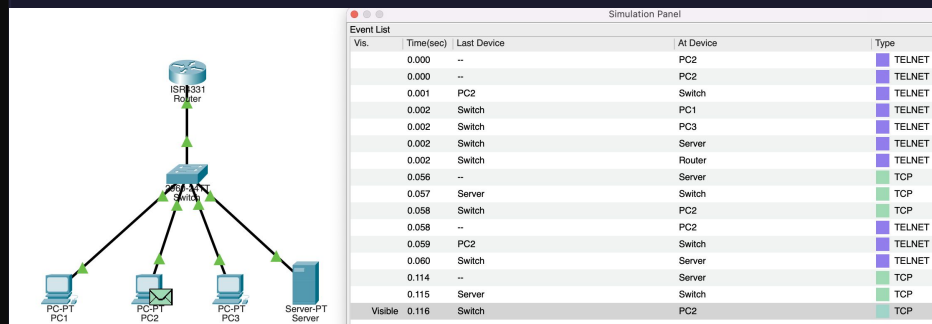
File transfer protocol

TCP Traffic Captured

Filters set to TCP + Telnet only. This isolates the port scanning traffic from background protocols like STP and CDP



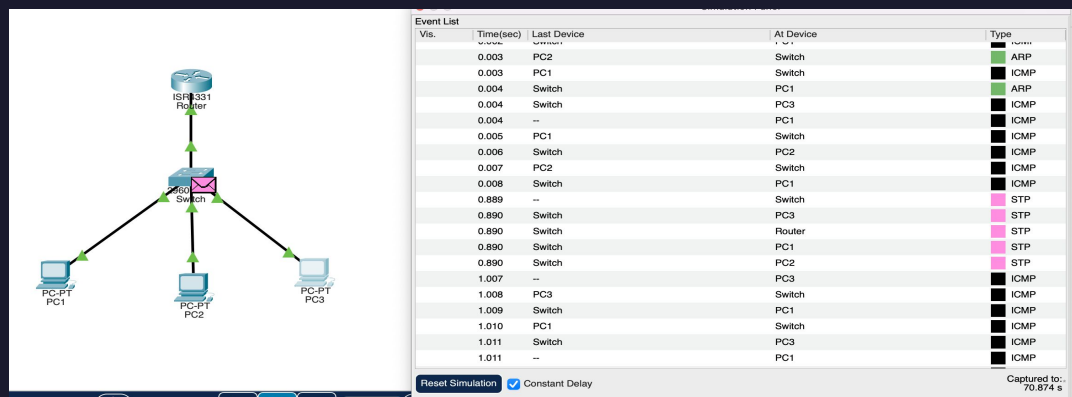
The Event List shows TELNET and TCP packets traveling between PC2, the Switch, and the Server, which is the same traffic pattern nmap produces during a port scan.



Each Telnet connection attempt to a different port number represents one port being checked, open ports respond, closed ports do not.

High-Volume Multi-Source Traffic

We ran simultaneous pings from PC1 and PC3 at the same time. This created traffic from two sources at once, producing a mixed-protocol spike that would be flagged as suspicious by a network monitoring tool



ICMP

Ping flood packets from PC1 and PC3 simultaneously: volume far exceeds normal single-source traffic.

ARP

Address resolution broadcasts triggered by new connections: indicates new traffic flows being initiated.

STP

Spanning Tree Protocol frames from the switch: normal background traffic mixed in with the anomaly.

MIXED

Multiple protocol types at once from multiple sources: This combination is the anomaly flag in monitoring tools.

Analysis & Security Threats

Normal Traffic

- Single ping with 4 packets, 0% packet loss
- ARP requests only when needed to resolve addresses
- Steady, low-volume traffic from one source at a time
- TCP connection to a single known port

Suspicious Traffic

- Rapid, repeated ICMP packets from one source (ping flood)
- Multiple Telnet connection attempts to different ports in sequence
- High-volume mixed ICMP, ARP and STP packets from multiple sources simultaneously
- TCP SYN packets sent to multiple ports probing for open services

Security Threats Identified

1 Ping Flood (DoS Attack)

A large number of ICMP packets are sent rapidly from one source to overwhelm a target and disrupt its availability.

2 Port Scanning (Reconnaissance)

Multiple connection attempts are made to different ports to identify open services and potential vulnerabilities.

3 High-Volume Multi-Source Traffic (Anomaly)

Simultaneous traffic from multiple devices creates an unusual spike in network activity that may indicate abnormal or suspicious behavior.

Real world Applications

The use of firewalls is a standard practice in almost every existing network since they help prevent unauthorized communication, and protect critical systems

Corporate Networks

Monitoring traffic logs to prevent misuse or information breaches.

Security teams inspect packets to differentiate between normal activity and suspicious patterns that indicate threats.

Web Server Safety

Verifying open ports (e.g., HTTP 80/HTTPS 443) to minimize attack surfaces.

Blocking unused ports is a critical standard practice to prevent unauthorized communication.

Higher Education

IT teams at universities analyze high-volume traffic patterns to detect hacked devices.

Identifying repeated port access attempts helps mitigate risks in large-scale student networks.

Challenges & Lessons

Challenges & Solutions

⚠ Wireshark not working properly

We experienced issues running Wireshark, likely due to firewall restrictions or software compatibility problems. To stay on track, we switched to Telnet, which was already familiar to some group members and allowed us to continue the lab efficiently.

⚠ Packets not visible in Simulation Mode

At first, we couldn't see the expected packets because the filters were not configured correctly. We resolved this by applying ARP and ICMP filters, which helped display the relevant network traffic.

⚠ Difficulty identifying attacks pings

It was challenging to distinguish attack pings from normal traffic. We addressed this by using TCP filtering in Telnet to reduce unnecessary noise and focus only on important packets.

Lessons Learned

01 — Importance of proper tool setup

Ensuring tools like Wireshark are correctly configured beforehand can save time and prevent delays.

02 — Effective use of filters

Applying the correct filters (ARP, ICMP, TCP) is essential for analyzing network traffic accurately.

03 — Flexibility in problem-solving

Switching to alternative tools like Telnet can help overcome unexpected technical issues.

04 — Improved packet analysis skills

We gained a better understanding of how to identify and differentiate network traffic, including potential attack patterns.

Conclusion

In this lab, we built and configured a network using Cisco Packet Tracer and used Simulation Mode to analyze network traffic. We examined packet details like IP addresses, MAC addresses and protocols. We stimulated different scenarios, including a ping flood attack, port scanning with Telnet, and mixed traffic from multiple sources. These helped us see how abnormal traffic patterns can appear in a network. Overall, this lab showed the importance of monitoring network activity and being able to recognize suspicious behavior. Understanding traffic at the packet level is important for improving network security.